Code No: 134BD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, April/May - 2023
FORMAL LANGUAGES AND AUTOMATA THEORY
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Define Kleene Closure and Positive Closure. [2]
b) Write about the applications of Finite Automata. [3]
c) Find the simplified regular expression for the following regular expression r (r*r + r*) + r*. [2]
d) Construct a regular grammar for L= $\{0^n1^n1/n>=1\}$. [3]
e) For the Grammar {S→AS/a, A→SbA/SS/ba} construct Left most derivation for the string aabbaaa? [2]
f) Define ambiguity in CFG with an example. [3]
g) What is the purpose of studying Turing Machine? [2]
h) Write about the programming techniques for Turing Machines. [3]
i) Define undecidability. Give an example of an undecidable problems. [2]
j) Compare recursive and recursive enumerable languages. [3]

PART – B

(50 Marks)

2. Construct NFA with ε which accepts a language consisting the strings of any number of 0's followed by any number of 1's followed by any number of 2's and also convert into NFA without ε transitions. [10]

OR

3.a) Design DFA for the following over {a,b}.
i) All strings containing not more than three a's.
ii) All strings that has at least two occurrences of b between any two occurrences of a.
b) Construct a DFA accepting the set of all strings ending with 00. [6+4]

4. Design a FA for the following languages:
a) $(0^*1^*)^*$
b) (0+1)* 111*
c) (0* 11* + 101). [3+3+4]

OR

5.a) Construct a DFA for the Regular Language consisting of any number of a's and b's.
b) Apply pumping lemma for the language L=$\{a^n/n$ is prime$\}$ and prove that it is not regular. [5+5]

6.a) Define Ambiguous Grammar. Check whether the grammar.
S→aAB, A→bC/cd, C→cd, B→c/d is ambiguous or not?
b) Show that for every PDA there exists a CFG such that L(G) = N(P). [5+5]

OR

7. Construct the CFG for the PDA M = ({q_0,q_1}, {0,1}, {R,Z_0}, δ, q_0, Z_0, Φ) and δ is givenby
δ(q_0,1,Z_0)=(q_0,RZ_0)
δ(q_0,1,R)=(q_0,RR)
δ(q_0,0,R)=(q_1,R)
δ(q_1,0, Z_0)=(q_0, Z_0)
δ(q_0,ε, Z_0)=(q_0,ε)
δ(q_1,1,R)=(q_1,ε). [10]

8.a) Design a Turing Machine for L={$0^n1^m0^n1^m$/m,n>=1}.
b) Define Chomsky Normal Form (CNF). Convert the following grammar to CNF
S→0S0|1S1| ε. [5+5]

OR

9.a) Explain following:
i) Closure properties of Context Free Languages.
ii) Decision properties of Context Free Languages.
b) Design a Turing machine to recognize all strings consisting of odd numbers of 1's. [5+5]

10.a) Discuss in brief about NP Hard problems.
b) Give the correspondence between P, NP and NP-complete problems. [5+5]

OR

11.a) Give an overview of recursively enumerable language.
b) Explain about the Decidability Problems. [6+4]

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